

Appendix C

Fundamentals of Mathematical Statistics

C.1 Populations, Parameters, and Random Sampling

Statistical inference involves learning something about a population given the availability of a sample from that population. By **population**, we mean any well-defined group of subjects, which could be individuals, firms, cities, or many other possibilities. By “learning,” we can mean several things, which are broadly divided into the categories of *estimation* and *hypothesis testing*.

A couple of examples may help you understand these terms. In the population of all working adults in the United States, labor economists are interested in learning about the return to education, as measured by the average percentage increase in earnings given another year of education. It would be impractical and costly to obtain information on earnings and education for the entire working population in the United States, but we can obtain data on a subset of the population. Using the data collected, a labor economist may report that his or her best estimate of the return to another year of education is 7.5%. This is an example of a *point estimate*. Or, she or he may report a range, such as “the return to education is between 5.6% and 9.4%.” This is an example of an *interval estimate*.

An urban economist might want to know whether neighborhood crime watch programs are associated with lower crime rates. After comparing crime rates of neighborhoods with and without such programs in a sample from the population, he or she can draw one of two conclusions: neighborhood watch programs do affect crime, or they do not. This example falls under the rubric of hypothesis testing.

The first step in statistical inference is to identify the population of interest. This may seem obvious, but it is important to be very specific. Once we have identified the population, we can specify a model for the population relationship of interest. Such models involve probability distributions or features of probability distributions, and these depend on unknown parameters. Parameters are simply constants that determine the directions and strengths of relationships among variables. In the labor economics example just presented, the parameter of interest is the return to education in the population.

Sampling

For reviewing statistical inference, we focus on the simplest possible setting. Let Y be a random variable representing a population with a probability density function $f(y;\theta)$, which depends on the single parameter θ . The probability density function (pdf) of Y is assumed to be known except for the value of θ ; different values of θ imply different population distributions, and therefore we are interested in the value of θ . If we can obtain certain kinds of samples from the population, then we can learn something about θ . The easiest sampling scheme to deal with is random sampling.

Random Sampling. If $Y_1, Y_2, \dots, Y_n$ are independent random variables with a common probability density function $f(y;\theta)$, then $\{Y_1, \dots, Y_n\}$ is said to be a **random sample** from $f(y;\theta)$ [or a random sample from the population represented by $f(y;\theta)$].

When $\{Y_1, \dots, Y_n\}$ is a random sample from the density $f(y;\theta)$, we also say that the Y_i are *independent, identically distributed* (or *i.i.d.*) random variables from $f(y;\theta)$. In some cases, we will not need to entirely specify what the common distribution is.

The random nature of $Y_1, Y_2, \dots, Y_n$ in the definition of random sampling reflects the fact that many different outcomes are possible before the sampling is actually carried out. For example, if family income is obtained for a sample of $n = 100$ families in the United States, the incomes we observe will usually differ for each different sample of 100 families. Once a sample is obtained, we have a set of numbers, say, $\{y_1, y_2, \dots, y_n\}$, which constitute the data that we work with. Whether or not it is appropriate to assume the sample came from a random sampling scheme requires knowledge about the actual sampling process.

Random samples from a Bernoulli distribution are often used to illustrate statistical concepts, and they also arise in empirical applications. If $Y_1, Y_2, \dots, Y_n$ are independent random variables and each is distributed as Bernoulli(θ), so that $P(Y_i = 1) = \theta$ and $P(Y_i = 0) = 1 - \theta$, then $\{Y_1, Y_2, \dots, Y_n\}$ constitutes a random sample from the Bernoulli(θ) distribution. As an illustration, consider the airline reservation example carried along in Appendix B. Each Y_i denotes whether customer i shows up for his or her reservation; $Y_i = 1$ if passenger i shows up, and $Y_i = 0$ otherwise. Here, θ is the probability that a randomly drawn person from the population of all people who make airline reservations shows up for his or her reservation.

For many other applications, random samples can be assumed to be drawn from a normal distribution. If $\{Y_1, \dots, Y_n\}$ is a random sample from the Normal(μ, σ^2) population, then the population is characterized by two parameters, the mean μ and the variance σ^2 . Primary interest usually lies in μ , but σ^2 is of interest in its own right because making inferences about μ often requires learning about σ^2 .

C.2 Finite Sample Properties of Estimators

In this section, we study what are called finite sample properties of estimators. The term “finite sample” comes from the fact that the properties hold for a sample of any size, no matter how large or small. Sometimes, these are called small sample properties. In Section C.3, we cover “asymptotic properties,” which have to do with the behavior of estimators as the sample size grows without bound.